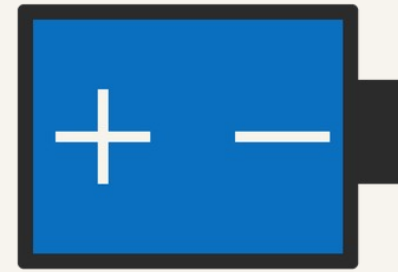


ELECTRICITY 2 · CURRENT, P.D., POWER &
RESISTANCE

I, V, P & R

The relationships behind every circuit calculation in this course.

Higher Physics



Starter

2008 Higher Physics

Scan the code and answer these:

Q15

Section A · uncertainty (measurements)

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

© Qualifications Scotland (SQA)

PART 1

Charge and current

Current is charge on the move — count how much passes, and how fast, and you have the relationship.

By the end of this part...

- 1 I can state what electric current is
- 2 I can use $Q = It$ to solve problems involving charge

Charge and current

- Electric current is the rate of flow of charge.
- Q , in coulombs (C), is the amount of charge.
- Current I , in amperes (A), is how much charge flows per second.

$$Q = It$$

$$Q = It$$

Q

charge (C)

I

current (A)

t

time (s)

1 coulomb is the charge that passes when 1 amp flows for 1 second.

Worked example – charge and current

A current of 3.0 A flows through a lamp for 40 s.
Find the charge that flows.

$$I = 3.0 \text{ A}$$

$$t = 40 \text{ s}$$

$$Q = ?$$

$$Q = It$$

$$Q = 3.0 \times 40$$

$$\underline{Q = 120 \text{ C}}$$

Your turn – charge and current

SHOW ME BOARDS

- 1 State what electric current measures.
- 2 A current of 0.50 A flows for 2.0 minutes. Find the charge that flows.
- 3 State the unit charge is measured in.

TRY THIS – Answers

- 1 The rate of flow of charge.
- 2 60 C (t must be converted to 120 s first).
- 3 The coulomb (C).

Starter

Revised Higher Specimen

Scan the code and answer these:

Q4

Section A · uncertainty (% uncertainties)

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

© Qualifications Scotland (SQA)

PART 2

Ohm's law and resistance

$V = IR$ links every quantity you can measure in a circuit to every quantity you can't see directly.

By the end of this part...

- 1 I can state what potential difference and resistance are
- 2 I can use $V = IR$ to solve circuit problems

Potential difference and resistance

- Potential difference (p.d.), V , is the energy transformed per coulomb of charge that passes between two points.
- Resistance, R , is how much a component opposes the flow of current.
- The three quantities V , I and R are linked by $V = IR$.

$$V = IR$$

$$V = IR$$

V

p.d. (V)

I

current (A)

R

resistance (Ω)

Rearrange freely: $I = V/R$ or $R = V/I$, depending what the question asks for.

Worked example – Ohm's law

A 12 V supply drives a current of 3.0 A through a resistor.
Find the resistance.

$$V = 12 \text{ V}$$

$$I = 3.0 \text{ A}$$

$$R = ?$$

$$R = V/I$$

$$R = 12 / 3.0$$

$$\underline{R = 4.0 \, \Omega}$$

Your turn – Ohm's law

SHOW ME BOARDS

- 1 State what potential difference measures.
- 2 A resistor of $5.0\ \Omega$ carries a current of $2.0\ \text{A}$. Find the p.d. across it.
- 3 State the unit resistance is measured in.

TRY THIS – Answers

- 1 The energy transformed per coulomb of charge between two points.
- 2 10 V.
- 3 The ohm (Ω).

Starter

2014 Higher Physics

Scan the code and answer these:



Section A · uncertainty (reaction time)

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

© Qualifications Scotland (SQA)

PART 3

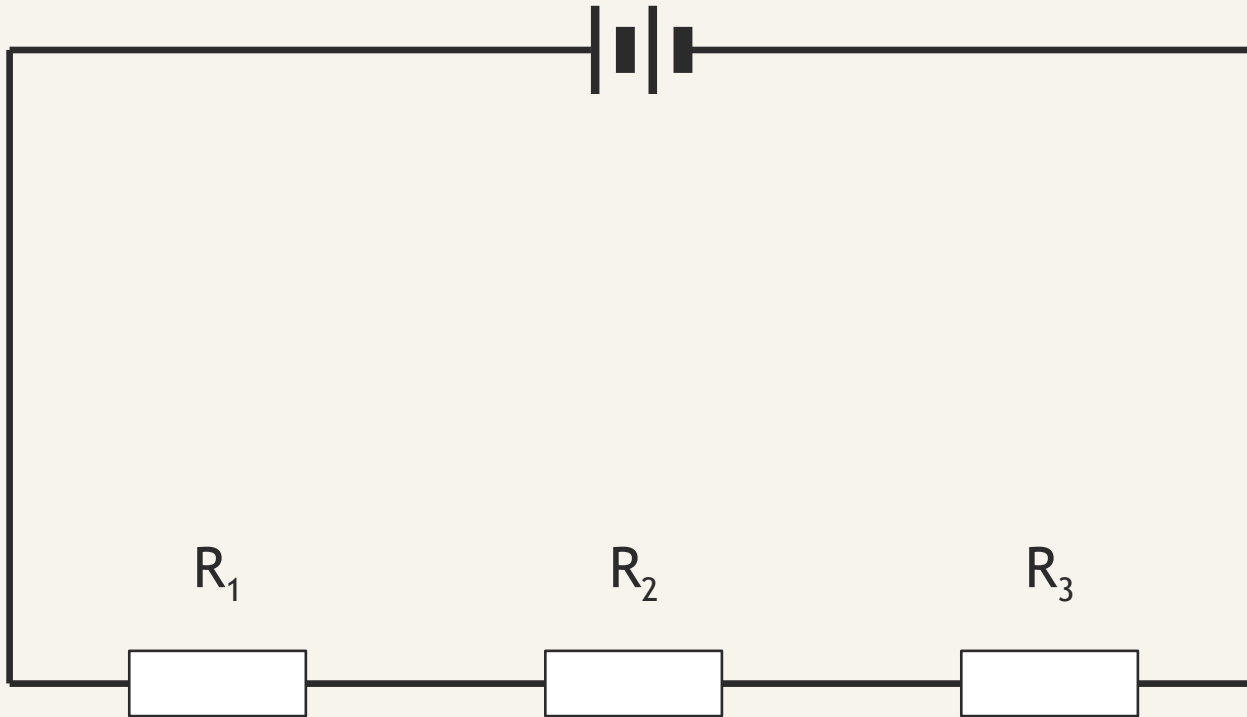
Series and parallel circuits

One path or many — the way resistors are wired changes how their resistances combine.

By the end of this part...

- 1 I can calculate total resistance for resistors in series
- 2 I can calculate total resistance for resistors in parallel

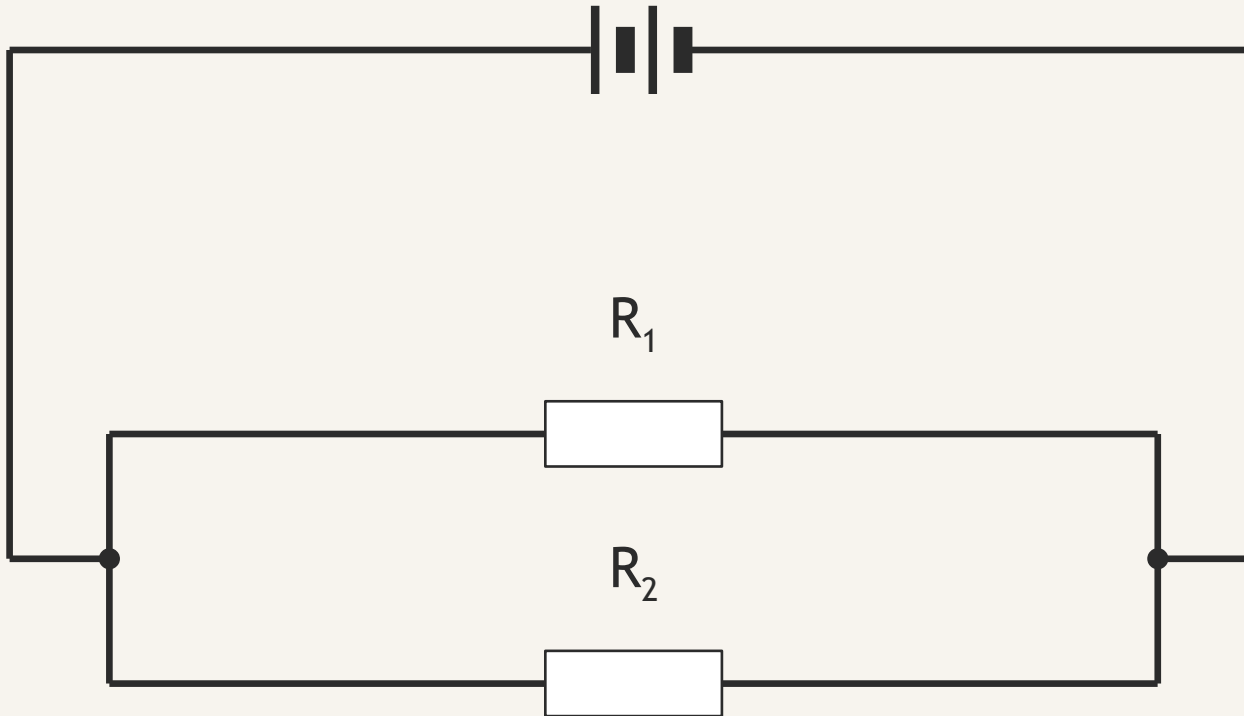
Resistors in series



$$R_T = R_1 + R_2 + \dots$$

One path — the charge meets every resistor in turn.

Resistors in parallel



$$1/R_T = 1/R_1 + 1/R_2 + \dots$$

More than one path —
total resistance is LESS
than the smallest branch.

Worked example – series resistance

Find the total resistance of a $10\ \Omega$, a $20\ \Omega$ and a $30\ \Omega$ resistor connected in series.

$$R_1 = 10\ \Omega$$

$$R_2 = 20\ \Omega$$

$$R_3 = 30\ \Omega$$

$$R_T = ?$$

$$R_T = R_1 + R_2 + \dots$$

$$R_T = 10 + 20 + 30$$

$$\underline{R_T = 60\ \Omega}$$

Worked example – parallel resistance

Find the total resistance of two $20\ \Omega$ resistors connected in parallel.

$$R_1 = 20\ \Omega$$

$$R_2 = 20\ \Omega$$

$$R_T = ?$$

$$1/R_T = 1/R_1 + 1/R_2 + \dots$$

$$1/R_T = 1/20 + 1/20 = 2/20$$

$$\underline{R_T = 10\ \Omega}$$

Watch Out – series vs parallel

THE MISTAKE

- Adding resistances directly for a parallel combination
- Forgetting parallel resistance is always less than any branch

THE CORRECT ANSWER

- Use $1/R_T$, not simple addition, in parallel.
- Two equal resistors in parallel halve, never double.

Your turn – series and parallel

SHOW ME BOARDS

- 1 Find the total resistance of a $5.0\ \Omega$ and a $15\ \Omega$ resistor in series.
- 2 Find the total resistance of a $6.0\ \Omega$ and a $3.0\ \Omega$ resistor in parallel.
- 3 State which combination – series or parallel – gives the greater total resistance.

TRY THIS – Answers

- 1 20 Ω .
- 2 2.0 Ω .
- 3 Series.

Starter

2015 Higher Physics

Scan the code and answer these:

Q16

Section A · uncertainty (wavelength)
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

© Qualifications Scotland (SQA)

PART 4

Electrical power

The same power relationship, written three ways — pick whichever form matches the two quantities you're given.

By the end of this part...

- 1 I can use $P = IV$, $P = I^2R$ and $P = V^2/R$ to solve power problems

Three forms of the power relationship

1 $P = IV$

2 $P = I^2R$

3 $P = V^2/R$



Electrical power

Power is the rate of transfer of energy.

$$P = IV$$

$$P = I^2 R$$

$$P = V^2 / R$$

Worked example – electrical power

A heater of resistance $15\ \Omega$ carries a current of $2.0\ \text{A}$.
Find the power dissipated.

$$I = 2.0\ \text{A}$$

$$R = 15\ \Omega$$

$$P = ?$$

$$P = I^2 R$$

$$P = 2.0^2 \times 15$$

$$\underline{P = 60\ \text{W}}$$

Watch Out – picking the right form

THE MISTAKE

- Using $P = IV$ when only V and R are given
- Squaring the wrong quantity in $P = I^2R$ or $P = V^2/R$

THE CORRECT ANSWER

- Use the form matching **ONLY** the values you have.
- Only the **SQUARED** quantity gets squared.

Your turn – electrical power

SHOW ME BOARDS

- 1 A 12 V supply drives a current of 0.50 A through a lamp. Find the power.
- 2 A 10 Ω resistor has 20 V across it. Find the power dissipated.
- 3 State which power relationship you'd use given only I and R.

TRY THIS – Answers

- 1 6.0 W.
- 2 40 W.
- 3 $P = I^2 R.$

Starter

2010 Higher Physics

Scan the code and answer these:

Q24

Section A · uncertainty (ball timing)
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

© Qualifications Scotland (SQA)

PART 5

Potential dividers

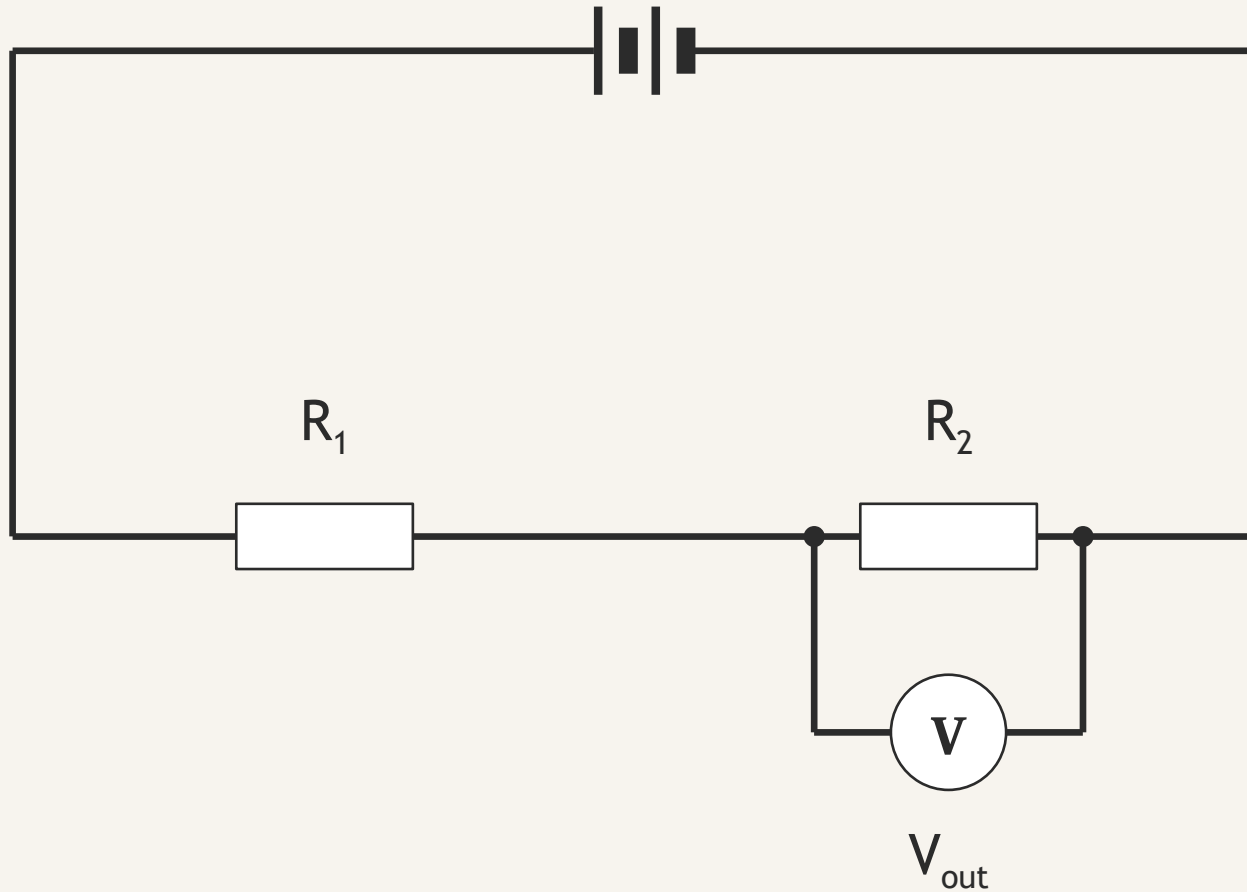
Two resistors in series share out a supply voltage in exact proportion to their resistances.

By the end of this part...

1

I can use the potential divider relationship to solve circuit problems

The potential divider



$$V_{out} = V_s R_2 / (R_1 + R_2)$$

V_{out} is measured across R_2 — the resistor the voltmeter is connected across.

Worked example – potential divider

A 12 V supply is connected across a 8.0 Ω and a 4.0 Ω resistor in series. Find the voltage across the 4.0 Ω resistor.

$$V_s = 12 \text{ V}$$

$$R_1 = 8.0 \text{ } \Omega$$

$$R_2 = 4.0 \text{ } \Omega$$

$$V_{\text{out}} = ?$$

$$V_{\text{out}} = V_s R_2 / (R_1 + R_2)$$

$$V_{\text{out}} = 12 \times (4.0 / 12)$$

$$\underline{V_{\text{out}} = 4.0 \text{ V}}$$

Watch Out – the potential divider ratio

THE MISTAKE

- Writing $V_1/V_2 = R_1^2/R_2^2$ instead of $V_1/V_2 = R_1/R_2$
- Using the wrong resistor on top of the divider fraction

THE CORRECT ANSWER

- The ratio is R_1/R_2 – never squared.
- V_{out} is measured across the top resistor, R_2 .

Your turn – potential dividers

SHOW ME BOARDS

- 1 A 10 V supply is split by a $3.0\ \Omega$ and a $2.0\ \Omega$ resistor in series. Find the voltage across the $2.0\ \Omega$ resistor.
- 2 State which resistor's voltage the potential divider formula gives when R_2 is on top of the fraction.
- 3 State what happens to V_{out} if R_2 is made much larger than R_1 .

TRY THIS – Answers

- 1 4.0 V.
- 2 The resistor whose value is on top – here, R_2 .
- 3 V_{out} approaches the full supply voltage.

Checkpoint

- 1 State the relationship between charge, current and time.
- 2 Use $V = IR$ to find the current through a $6.0\ \Omega$ resistor with $18\ \text{V}$ across it.
- 3 State the rule for combining resistors in series, and in parallel.
- 4 Write the three forms of the electrical power relationship.
- 5 State the potential divider relationship, and what it gives you.

Exam Practice

2025 Q19 – total resistance of a network.

2025 Q20 – power in a lamp.

2024/2025 S3 – potential divider ratios
(subscripts, not powers).

Past papers, Section A

LOOKING AHEAD

Next: Electrical Sources

What happens when a real battery isn't quite the perfect supply this topic has assumed so far.

Plenary

- 1 State the relationship between charge, current and time.
- 2 State Ohm's law.
- 3 State how total resistance is found for resistors in series, and in parallel.
- 4 Write the three forms of the power relationship.
- 5 State the potential divider relationship.

- 1 Explain why adding a resistor in parallel always reduces total resistance.
- 2 Two potential dividers share the same supply. Explain what changes if R_1 and R_2 are swapped.

Extension

EXTRA PRACTICE

- 1 Derive $P = V^2 / R$ by combining $P = IV$ and $V = IR$.
- 2 A potential divider has $R_1 = R_2$. Explain, without calculating, why V_{out} is exactly half the supply voltage.

Extension – EXTRA

EXTRA PRACTICE

- 1 Research how a potential divider with a thermistor or LDR is used as a sensor circuit.